

Smart Medicine Reminder To Improve Medication Adherence By Using Medi-Alert IoT

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Abstract—In today's fast-paced world, many individuals, especially elderly or chronically ill patients, often forget to take their prescribed medicines on time. To tackle this issue, a Smart Medicine Reminder System has been created using Arduino UNO, Real Time Clock (RTC DS1307), an LCD display, and a buzzer alert. The system continuously monitors real-time data from the RTC and compares it with set medicine times. When the time matches, the buzzer goes off, and a reminder message appears on the LCD screen, prompting the user to take their medicine. A push button lets the user acknowledge the alert and stop the buzzer. This paper aims to improve medication adherence, reduce human error, and offer a cost-effective solution for healthcare assistance. The system was designed, simulated, and tested using Proteus 8 Professional and programmed with Arduino IDE. Results show accurate timekeeping, reliable alerts, and ease of use, making it suitable for both home and hospital settings and overall patient health.

Keywords — Medication Adherence, Smart Pill Box, GSM Alert System, Elderly Healthcare, Remote Monitoring.

I. INTRODUCTION

Technology is now a key part of our daily lives. It has significantly raised living standards and increased individual needs. One of these needs is for better healthcare solutions, especially for those who take daily medications. A large number of people around the world rely on prescribed medications. As the number of prescriptions grows, so does the risk of missing doses or taking the wrong amounts.

The American Society of Health-System Pharmacists reports that nearly one-third of older adults take eight or more medications each day. Managing so many medicines raises the chances of missing doses or, even worse, taking the same medication more than once. The British Pharmacological Society found that over 80% of elderly hospital admissions are due to harmful drug reactions, many of which come from dosage mistakes.

Prescription bottles often have detailed instructions, but these can be unhelpful for individuals with memory issues like Alzheimer's disease. A United Nations report on Ageing and Development from 2009 notes that the number of

people aged 60 and older is rising quickly. Many older adults live alone or deal with physical or mental challenges, making self-care harder. In these cases, delays or mistakes in taking medications can lead to serious health problems.

In addition, caregivers and nurses often have a tough time keeping track of the remaining pills and medication schedules, which can create more complications. This growing problem underscores the urgent need for smart healthcare systems that help elderly individuals manage their medications better. Devices that track pill usage, send real-time alerts, monitor vital signs, or notify healthcare professionals during emergencies can significantly improve the safety and quality of life for older adults.

Smart healthcare devices can significantly improve quality of life. They can detect vital signs, send emergency alerts, or monitor medication use, offering timely help to both patients and caregivers.

II. LITERATURE SURVEY

Several researchers have proposed systems to improve medication adherence using smart pill boxes and IoT-based reminders. This section reviews key contributions in the field, highlighting their methods, features, and limitations.

[1] The base paper for this project is titled "IoT-Based Real-Time Medicine Reminder and Dispenser for Elderly Patients", published in the 2024 2nd World Conference on Communication & Computing (WCONF), Raipur, India, and presented by Safwan Sulaiman Sadi, Abdul Awal Tanvir, Arafath Al Hossain, Mohammad Faisal Uddin, Mohammad Rejwan Uddin, and Mahady Hasan from Independent University, Bangladesh.

This paper focuses on solving the common problem of medication non-adherence among elderly patients using IoT technology. The proposed system is an IoT-enabled smart medicine dispenser that

automatically reminds users to take their prescribed medicines at the correct time through four types of alerts — an LCD display, a speaker announcement, a mobile notification, and an automatic dispensing mechanism.

The system is built using a NodeMCU (ESP8266) microcontroller, integrated with an RTC module (DS3231), servo motor, buzzer, and LCD display. The device connects to the internet to synchronize real-time schedules and notify the patient via mobile devices. The paper also includes simulation results, system topology, algorithm design, hardware testing. [2]“IoT Based Fruit Quality Inspection and Lifespan Detection System”. The fruit quality inspection and lifespan detection system based on IoT demonstrated in this study is an efficient fruit and vegetable quality monitoring solution. The system detects the quality of fruit in real-time by using sensors to monitor the parameters like temperature, humidity, and gas emissions. The data is sent to a cloud server for surveillance from a distance. The paper is aimed at reducing food waste, bettering storage management, and helping the suppliers in keeping the much-needed freshness during transit. [3] The Unmet Challenge of Medication Nonadherence This paper discusses the prevalent problem of medication nonadherence where patients do not take their medications as prescribed. The paper cites reasons like forgetfulness, misunderstanding, high price, and side effects. The author points out that the end result of nonadherence is that it becomes a double-edged sword in terms of treatment; it makes the treatment fail and also takes a toll on healthcare expenses. Healthcare practices would be the backbone in supporting adherence by means of good communication, taking care of the patient's needs, and using technology to provide reminders, according to the research.

[4] Hossain et al. (2023) – Nursing Robot for Non-Invasive Health Monitoring In this study, a nursing robot that can take human body temperature and heart rate readings without contact is introduced. The robot consists of sensors, IoT, and AI which provide it with the capability of data collection and analysis in real-time. It is a solution created specifically for the health professionals in hospitals that will relieve them from the burden of manual monitoring and also provide the patients with the safety and the continuous remote observation of their vital signs.[5] O’Neill et al. (2023) – Medicinal

Cannabis and Implications for Workplace Health and Safety The authors of this scoping review evaluated various systematic reviews in order to uncover the possible dangers and difficulties of using medicinal cannabis in the workplace. They came to the conclusion that while medicinal cannabis does have therapeutic properties, it could also lead to cognitive and motor function impairment, which in turn, would negatively affect the performance and safety of the employees. The authors recommended clear policies for the workplace, training of staff, and the provision of regulatory guidelines as a way to ensure both the patient's safety and the safety of the workplace. The review finally reached the conclusion that future studies and operational management strategies are the pre-requisites for the coexistence of therapeutic use and occupational safety.

[6] Choudhry et al. (2017) – Effect of Reminder Devices on Medication Adherence (REMIND Trial) A randomized clinical trial was conducted to find out whether simple reminder devices would be able to significantly enhance the medication adherence of patients with chronic diseases. The patients were given the opportunity of using different types of reminder devices, such as the electronic pill boxes and digital alerts. The researchers came to the conclusion that the use of reminder devices alone led to only a tiny increase in adherence rates. The authors also noted that in addition to the devices, behavioural, motivational, and educational interventions are also crucial if one aims at achieving significant adherence improvements. Thus, the research has provided strong support to the use of integrated reminder and monitoring systems in the case of patients that have to do with long-term drug therapies. [7] Kader et al. (2018) – Automated Reminder Medicine Box for Elderly and Hospitals This paper describes the design and realization of an automated medicines box that supports seniors and hospitals in the proper management of medication schedules. The machine's operation includes the use of microcontrollers, real-time clocks, and alarm systems to notify patients of their specific medication times.

III. METHODOLOGY

The Smart Medicine Reminder System helps patients take their medicines on time. It uses an Arduino-based embedded system with a Real Time Clock (RTC), an LCD display, a buzzer, and control buttons. The system monitors data and alerts the user when it is time to take the prescribed medicine.

The DS1307 RTC module tracks the current time even when the system is off. The Arduino UNO reads the time at regular intervals from the RTC via I²C communication (SDA, SCL). The program compares the current time with scheduled medicine times (for example, 08:00, 14:00, and 20:00). When the current time matches a medicine schedule. The buzzer is activated with a BC547 transistor as a driver circuit. The LCD displays a message like “Take Medicine Now. The user can press the STOP button to silence the buzzer and reset the system to normal display mode. The cycle continues throughout the day, ensuring that the patient receives reminders at the correct times.

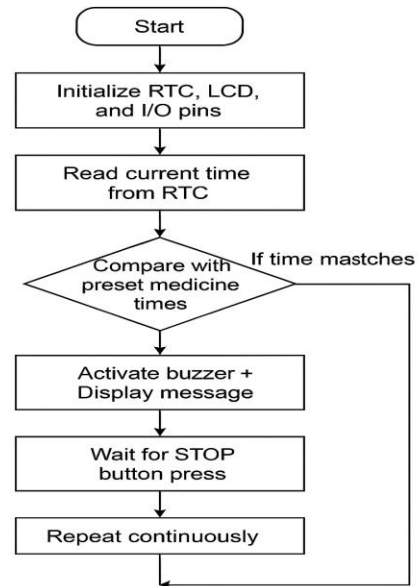
IV. IMPLEMENTATION STEPS

1. **Circuit Design:** The circuit was simulated in Proteus 8 Professional using Arduino UNO, DS1307 RTC, the LCD module, a buzzer, and push buttons.
2. **Coding:** An embedded C program was written in the Arduino IDE and uploaded to the microcontroller. The code manages RTC communication, time display, and alarm triggering.
3. **Simulation:** The entire circuit was tested in Proteus. The buzzer and LCD responses were checked at the scheduled times.
4. **Testing:** Functionality was confirmed by manually changing the RTC time and ensuring the alarms triggered correctly.

V. FLOW OF OPERATION

The flowchart represents the working process of a Medicine Reminder System using an RTC module. It starts by initializing the RTC, LCD, and input/output pins, then continuously reads the current time from the RTC. The system compares this time with preset medicine times, and if a match occurs, it activates a buzzer and displays a reminder message on the LCD. It then waits for the user to press the STOP button to stop the buzzer and clear the display. This process repeats continuously to

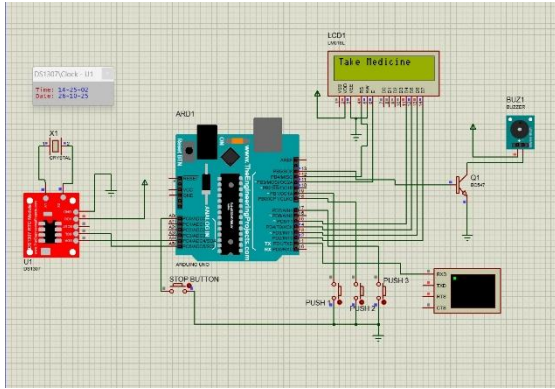
ensure timely alerts for every scheduled medicine dose.



VI. RESULT

The image provided showcases the Proteus simulation of a Smart Medicine Reminder System developed using an Arduino Uno microcontroller. This project is designed to help patients, particularly elderly individuals, remember to take their medications on time. The system includes several essential components such as an Arduino Uno, a DS1307 Real-Time Clock (RTC) module, a 16x2 LCD display (LM016L), a buzzer (BUZ1), a BC547 transistor, and various push buttons for user interaction. The DS1307 RTC module keeps track of the accurate time and date even during power outages, thanks to its internal battery backup.

The Arduino Uno acts as the central control unit, processing real-time data from the RTC and comparing it with the preset medication times stored in the program. When the current time matches the scheduled reminder for medication, the Arduino sends a signal to the buzzer circuit via the BC547 transistor, triggering an audible alert to notify the user. At the same time, the LCD display presents a reminder message like “Take Medicine” for visual reinforcement.



The push buttons connected to the Arduino allow users to set and adjust the time, schedule medication alarms, and stop the buzzer once they have taken their medicine. Additionally, there is a serial terminal included in the circuit for debugging and communication purposes. This simulation illustrates how an effective IoT-based healthcare reminder system works to ensure that patients do not miss their doses. The combination of hardware components like the RTC and buzzer with the Arduino enhances both reliability and ease of use. Overall, this Smart Medicine Reminder System fosters better health management and timely medication adherence while helping patients efficiently maintain their treatment schedules through real-time monitoring, alert systems, and straightforward user interaction.

CONCLUSION

The proposed solution involves developing a GSM-based automatic pill dispenser. This system integrates technologies to improve medication adherence, especially among elderly or chronically ill patients. It uses a Real-Time Clock (RTC) to schedule and automatically dispense medication at set intervals. Pills can be loaded daily or weekly, making the system flexible and user-friendly.

One key feature is its ability to send SMS alerts to caregivers if a dose is missed. This helps ensure safety and timely intervention. The device's user interface is designed to be easy to use, even for elderly patients. The system also allows for modular expansion, enabling users to add more compartments or increase pill capacity as needed. By combining automation with GSM communication, this Smart Medicine Box (SMB) significantly improves treatment effectiveness, supports independent living, and reduces the risk of serious incidents caused by missed medication.

FUTURE SCOPE

Since the introduction of pills in the medication industry, there has been a need for a device that can deliver medicine in place of a person. No pill dispenser today can determine who should receive a pill, when to give it, or how many pills to distribute. Home pill users face various issues, such as someone stealing pills, forgetting to take them, or having too many pills available at once. The AMDR mechanism operates on the principle of mechatronics, combining mechanics and electronics to dispense medicine. As a result, the chances of error are significantly reduced.

The following sectors can benefit from the AMDR:

- i. Hospitals: The automatic pill dispenser can streamline the pill dispensing process in both large and small healthcare centers. With the help of AMDR, medical staff and doctors can save time and reduce the need for frequent check-ups on every patient. This allows doctors to focus more on patients in critical condition or those who require closer attention from medical staff.
- ii. Home: Vitamins and other essential dietary supplements needed daily before and after meals represent a significant opportunity for automatic pill dispensers. An individual keychain can trigger pill dispensing for the user's personal diet, and a buzzer can remind them when to take their supplements.
- iii. Elderly: The automatic pill dispenser offers an excellent solution for elderly individuals to remember when to take their pills. The access key can be worn around the neck or as a bracelet for convenience.

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